Question ID: 07bcecac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

$$P(t) = 24.8(1.036)^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement “$P(8)$ is approximately equal to 32.91”?

Answer

A. In 1984, the predicted population of this country was approximately 8 million.

B. In 1984, the predicted population of this country was approximately 32.91 million.

C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.

D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

Correct Answer: C

Rationale

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P(8)$ is the value of $P(t)$ when $t = 8$, it follows that "$P(8)$ is approximately equal to 32.91" means that the value of $P(t)$ is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement "$P(8)$ is approximately equal to 32.91" is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.8(1.036)^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.